

A Sharp Critical Disk Case and Attainment in the Fractional Morrey Branch

Run fa_banach_001 — lane 17

12 August 2026

Status: partial new result plus literature answer. For the critical case $(n, s, p) = (2, 1, 2)$, the sharp dimensionless constant in the source’s whole-space Poincaré–Wirtinger inequality on a disk is $j_{0,1}^{-2}$, and all extremals are explicit. Separately, arXiv:2309.06058 proves attainment for the source’s entire supercritical fractional Morrey branch. The general sharp-constant problem remains open.

1 Source question and decomposition into branches

Brasco, Gómez-Castro, and Vázquez characterize homogeneous fractional Sobolev spaces in arXiv:2007.08000v2 [1]. Section 6 on arXiv PDF page 22 asks:

Source question (Section 6). *What are the optimal constants, and are they achieved by extremal functions, in the subcritical fractional Sobolev inequality, the critical Poincaré–Wirtinger inequality, and the supercritical fractional Morrey inequality?*

22

BRASCO, GOMEZ-CASTRO, AND VAZQUEZ

6. COMMENTS AND OPEN QUESTIONS

1. Our three embedding theorems pose the question of what are the optimal constants, and whether they are achieved by extremal functions. We specifically refer to inequality (3.1) for $sp < n$, inequality (2.9) for $sp = n$, and inequality (2.11) for $sp > n$.

Note that in the case $s = 1$, the extremals in the Gagliardo–Nirenberg–Sobolev range $1 < p < n$ were found by Aubin [2] and Talenti [28]. The extremals in the Morrey range $p > n$, still for $s = 1$, have been recently described by Hynd and Sauffert [15]. We do not know of any similar result in the limit case $p = n$.

2. In the case of proper subsets $\Omega \subset \mathbb{R}^n$ the characterization result will depend on the different options.

The question is a program rather than a single yes/no assertion. The source itself recalls that subcritical extremals are already studied and that the sharp subcritical constant is explicit for $p = 2$ but unknown for general p . Below we obtain an exact critical case, then record a later answer to the Morrey-attainment branch.

2 New result: the sharp critical constant on a planar disk

Write $j = j_{0,1} = 2.404825557\dots$ for the first positive zero of the Bessel function J_0 .

Theorem 1. For every $u \in C_c^\infty(\mathbb{R}^2)$, every $x_0 \in \mathbb{R}^2$, and $R > 0$,

$$\int_{B_R(x_0)} |u - u_{x_0,R}|^2 dx \leq \frac{R^2}{j_{0,1}^2} \int_{\mathbb{R}^2} |\nabla u|^2 dx. \quad (1)$$

The constant $j_{0,1}^{-2} = 0.1729150690\dots$ is sharp. It is attained in the homogeneous completion $\mathcal{D}^{1,2}(\mathbb{R}^2)$ modulo constants.

For $B_1(0)$, every nonconstant extremal is, for some $(A, B) \neq (0, 0)$ and $c \in \mathbb{R}$,

$$U(r, \theta) = c + \begin{cases} J_1(jr)(A \cos \theta + B \sin \theta), & 0 \leq r \leq 1, \\ J_1(j)r^{-1}(A \cos \theta + B \sin \theta), & r \geq 1. \end{cases} \quad (2)$$

The extremals for other disks follow by translation and scaling.

Proof intuition

The source uses the whole-space Dirichlet energy, not merely the energy inside the disk. For a prescribed boundary Fourier mode, minimize the exterior energy first. The mode of angular frequency $m \geq 1$ continues as r^{-m} and contributes exactly a boundary penalty $m|f(1)|^2$. The resulting first interior mode has a Robin condition. Its Bessel equation reduces that condition to $J_0(j) = 0$, which is lower than both the radial mean-zero mode and all higher angular modes.

Proof. By translation and scaling it is enough to prove the assertion for $B = B_1(0)$. Since both sides ignore additive constants, replace u by $u - u_B$. Thus $u_B = 0$.

Expand in polar Fourier modes. Apart from the harmless normalization factors 2π for the radial mode and π for the other modes, the denominator is the sum of

$$\int_0^1 |f(r)|^2 r dr,$$

while the Dirichlet energy of an angular mode $m \geq 1$ is

$$\int_0^\infty \left(|f'|^2 + \frac{m^2}{r^2} |f|^2 \right) r dr.$$

For the exterior part, completing a square gives

$$\begin{aligned} \int_1^\infty \left(|f'|^2 + \frac{m^2}{r^2} |f|^2 \right) r dr &= m|f(1)|^2 + \int_1^\infty \left| f' + \frac{m}{r} f \right|^2 r dr \\ &\geq m|f(1)|^2. \end{aligned} \quad (3)$$

Equality holds exactly for the finite-energy harmonic continuation $f(r) = f(1)r^{-m}$.

It follows that the least Rayleigh quotient in mode $m \geq 1$ is bounded below by the first eigenvalue of

$$Q_m(f) = \int_0^1 \left(|f'|^2 + \frac{m^2}{r^2} |f|^2 \right) r dr + m|f(1)|^2. \quad (4)$$

Since $Q_m(f) \geq Q_1(f)$ for every $m \geq 1$, it suffices to solve the first mode. Its Euler–Lagrange equation and natural boundary condition are

$$-(rf')' + \frac{1}{r}f = \lambda rf, \quad f'(1) + f(1) = 0,$$

with the regular condition at 0. Thus $f(r) = J_1(\sqrt{\lambda}, r)$, and

$$\sqrt{\lambda}J_1'(\sqrt{\lambda}) + J_1(\sqrt{\lambda}) = \sqrt{\lambda}J_0(\sqrt{\lambda}) = 0.$$

The first eigenvalue is therefore $j_{0,1}^2$.

For the radial mode, the mean-zero condition reads $\int_0^1 f(r)r dr = 0$. The exterior energy is nonnegative, and the radial Neumann Poincaré inequality gives

$$\int_0^1 |f'|^2 r dr \geq j_{1,1}^2 \int_0^1 |f|^2 r dr.$$

The standard interlacing of Bessel zeros gives $j_{1,1} > j_{0,1}$. Hence the radial mode is strictly more expensive. Higher modes are no cheaper by $Q_m \geq Q_1$. Summing all modes proves (1) for $R = 1$.

Equality forces the first angular mode, the first Robin eigenfunction, and equality in (3). These conditions give exactly (2). The identity $jJ_1'(j) + J_1(j) = jJ_0(j) = 0$ makes the radial derivative continuous at $r = 1$, so U is a global weak extremal. It has finite Dirichlet energy. Truncating its r^{-1} tail and then mollifying approximates it by C_c^∞ functions in Dirichlet energy and locally in L^2 . Therefore the constant is sharp already for the source's test class, and attained in its homogeneous completion. Scaling gives the factor R^2 for a general disk. $\square$

3 Upgrade attempts and the remaining obstruction

The compactness part actually extends beyond the Hilbertian disk: if $sp = n$ with $s < n$, the critical homogeneous quotient is reflexive and its restriction to a fixed ball, normalized to have mean zero, is compact in L^p by local Rellich compactness. Hence the sharp critical Poincaré–Wirtinger quotient is attained in the homogeneous completion. This gives attainment, but not a usable closed form for the constant.

The exact upgrade above works because $(n, s, p) = (2, 1, 2)$ is simultaneously local, linear, and rotationally separable. For local $p = n \neq 2$, the angular decomposition does not diagonalize the nonlinear p -energy. In the other Hilbertian critical case $(n, s, p) = (1, 1/2, 2)$, restriction to an interval leads to a nonlocal exterior Dirichlet-to-Neumann operator rather than the scalar penalty in (3). These are structural obstructions to turning the present calculation into a full all-parameter formula, not merely missing algebra.

4 Later literature: Morrey extremals do exist

Brasco, Prinari, and Sk explicitly cite the source paper and study the sharp fractional Morrey constant in arXiv:2309.06058v1 [2]. Their Theorem 6.1 on arXiv PDF page 23 proves, for $N \geq 1$, $0 < s \leq 1$, $1 < p < \infty$ and $sp > N$, that the sharp variational infimum over the homogeneous Morrey space is attained.

6. EXISTENCE OF EXTREMALS

We still denote by $\mathfrak{m}_{s,p}(\mathbb{R}^N)$ the sharp constant defined in (1.2).

Theorem 6.1. *Let $N \geq 1$, $0 < s \leq 1$ and $1 < p < \infty$ be such that $sp > N$. Then we have*

$$\mathfrak{m}_{s,p}(\mathbb{R}^N) = \inf_{\varphi \in \mathcal{W}^{s,p}(\mathbb{R}^N)} \left\{ [\varphi]_{W^{s,p}(\mathbb{R}^N)}^p : [\varphi]_{C^{0,\alpha_{s,p}}(\mathbb{R}^N)} = 1 \right\}.$$

Moreover, the infimum on the right-hand side is attained.

Proof. Since $C_0^\infty(\mathbb{R}^N) \subseteq \mathcal{W}^{s,p}(\mathbb{R}^N)$, we immediately obtain

$$\mathfrak{m}_{s,p}(\mathbb{R}^N) \geq \inf_{\varphi \in \mathcal{W}^{s,p}(\mathbb{R}^N)} \left\{ [\varphi]_{W^{s,p}(\mathbb{R}^N)}^p : [\varphi]_{C^{0,\alpha_{s,p}}(\mathbb{R}^N)} = 1 \right\}.$$

For the reverse inequality, we observe at first that

Thus the attainment half of the source's supercritical branch has a positive literature answer. The later paper explicitly says that the numerical value of the sharp constant remains unknown, already in much of the local theory; it supplies existence, regularity, and sharp asymptotics rather than a closed formula.

5 Novelty check and scope

The four lightweight run indexes contained no treatment of either arXiv id. A bounded exact-title, formula, Beppo–Levi, disk, Bessel-zero, and Poincaré–Wirtinger search through 12 August 2026 found the source, the later Morrey papers, standard bounded-domain Poincaré constants, and general exterior-energy references, but no source recording the whole-space disk constant $j_{0,1}^{-2}$ or the extremals (2). The proof is short enough to audit directly, but bibliographic novelty remains subject to human review.

This packet does not claim the full three-regime program. It gives a complete sharp answer for one critical parameter choice, a general critical attainment argument without an explicit constant, and the exact later provenance for Morrey attainment. The general nonlinear/fractional constants remain open.

References

- [1] L. Brasco, D. Gómez-Castro, and J. L. Vázquez, *Characterisation of homogeneous fractional Sobolev spaces*, Calc. Var. Partial Differential Equations **60** (2021), Paper 60; arXiv:2007.08000v2.
- [2] L. Brasco, F. Prinari, and F. Sk, *On Morrey's inequality in Sobolev–Slobodeckii spaces*, arXiv:2309.06058v1 (2023).